

Weekly Review and Practice Practical (2 hours): Create a network with multiple VLANs and configure inter-VLAN routing. Task: Submit a report on VLAN and inter-VLAN routing setup.

Q-1 On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

[A]

- **Create a network containing:**
 - 3 switches
 - 6 PCs — 2 PCs connected to each switch
 - VLAN 10 → **Sales**
 - VLAN 20 → **Support**
- **Device connections**

Device	Port	Connected To	VLAN
PC1	NIC	SW1 Fa0/1	VLAN 10 Sales
PC2	NIC	SW1 Fa0/2	VLAN 20 Support
PC3	NIC	SW2 Fa0/1	VLAN 10 Sales
PC4	NIC	SW2 Fa0/2	VLAN 20 Support
PC5	NIC	SW3 Fa0/1	VLAN 10 Sales
PC6	NIC	SW3 Fa0/2	VLAN 20 Support
SW1	Fa0/23	SW2 Fa0/23	Trunk
SW2	Fa0/24	SW3 Fa0/23	Trunk
R1	G0/0	SW1 Fa0/24	Trunk

➤ **IP Addressing**

VLAN 10 — Sales

Network: 192.168.10.0/24

Gateway:

192.168.10.1

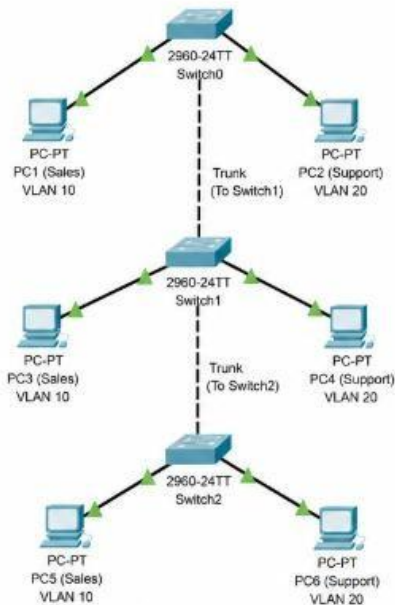
PC	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC3	192.168.10.12	255.255.255.0	192.168.10.1
PC5	192.168.10.13	255.255.255.0	192.168.10.1

VLAN 20 — Support Network:

192.168.20.0/24 Gateway:

192.168.20.1

PC	IP Address	Subnet Mask	Default Gateway
PC2	192.168.20.11	255.255.255.0	192.168.20.1
PC4	192.168.20.12	255.255.255.0	192.168.20.1
PC6	192.168.20.13	255.255.255.0	192.168.20.1



Q-2 Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

[A]

SW1 → SW2 trunk On

SW1: enable

configure terminal

interface fa0/23 switchport

mode trunk

switchport trunk allowed vlan 10,20

exit

interface fa0/24 switchport

mode trunk

switchport trunk allowed vlan 10,20

exit

end write

memory

Here:

- Fa0/23 → SW2
- Fa0/24 → Router R1

SW2 → SW1 and SW3 trunks

On **SW2**: enable

configure terminal

interface fa0/23 switchport

mode trunk switchport

trunk allowed vlan 10,20

exit

interface fa0/24 switchport mode

trunk switchport trunk allowed vlan

10,20 exit

end write

memory

Here:

- Fa0/23 → SW1
- Fa0/24 → SW3

SW3 → SW2 trunk On

SW3: enable

configure terminal

```
interface fa0/23 switchport mode
trunk switchport trunk allowed vlan
10,20 exit

end write
memory
```

Verify Trunking

Run this command on each switch:

```
show interfaces trunk
```

Q-3 Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

[A]

Router configuration Connect:

R1 G0/0 ↔ SW1 Fa0/24

Open **R1** → **CLI**.

First configure the physical interface:

```
enable configure
terminal
```

```
interface gigabitEthernet 0/0 no
shutdown
exit
```

Now configure VLAN 10: interface

```
gigabitEthernet 0/0.10
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
exit
Configure VLAN 20: interface
gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit Finally: end write memory
```

Router Configuration — Complete Version

You can also enter everything together:

```
enable configure
terminal

interface gigabitEthernet 0/0 no
shutdown
exit

interface gigabitEthernet 0/0.10
encapsulation dot1Q 10 ip address
192.168.10.1 255.255.255.0 exit

interface gigabitEthernet 0/0.20
encapsulation dot1Q 20 ip address
192.168.20.1 255.255.255.0 exit

end write
memory
```

Verify Router Configuration Run:

show ip interface brief You
can also check:
show running-config

Configure PC IP Addresses

PC → Desktop → IP Configuration

PC1

IP Address: 192.168.10.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

PC2

IP Address: 192.168.20.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

PC3

IP Address: 192.168.10.12
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

PC4

IP Address: 192.168.20.12
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

PC5

IP Address: 192.168.10.13
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

PC6

IP Address: 192.168.20.13

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

Q-4 Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works. Hint : Check subinterface IPs, trunk configuration, and default gateways on PCs.

[A]

PC1 → PC2

PC1 belongs to VLAN 10:

192.168.10.11

PC2 belongs to VLAN 20:

192.168.20.11

PC1 :

Desktop → Command Prompt

Enter:

ping 192.168.20.11 Expected

result:

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127 Therefore,

inter-VLAN communication is successful.



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```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=126
Reply from 192.168.20.11: bytes=32 time<1ms TTL=126
Reply from 192.168.20.11: bytes=32 time<1ms TTL=126
Reply from 192.168.20.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```